

Sparse Concept Channels in Frozen 3D CT Vision Encoders

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Abstract

Large vision-language models are becoming increasingly dominant in 3D medical image interpretation, but we rarely know *which* internal units encode clinical findings or *where* that information lives in the representation. We first study this on a 3D chest vision-language model (PILLAR-0) by probing its frozen vision embeddings. We show that (i) each radiological finding is encoded by a *sparse* set of ~ 10 vision-encoder channels that match full-feature classification performance and far exceed a zero-shot text prompting; (ii) turning off the channels tied to one finding, that finding’s score collapses while unrelated labels stay stable; and (iii) the same sparse probe *replicates* on an architecturally unrelated 3D abdominal VLM (MERLIN) suggesting a general property of frozen medical encoders. Our training-free concept channel probe (CCP) method, paired with a corpus-derived report template, outperforms published CT-CHAT on clinical efficacy and NLG metrics (F1 0.549 vs. 0.184; BLEU 0.483 vs. 0.373) at $22\times$ lower latency. Our results provide a clear, reproducible characterization of how frozen medical encoders represent findings, demonstrating direct applicability across models.

Introduction

Foundation models for medical imaging are now strong enough to be considered as clinical decision support tools, yet their internal representations are poorly understood. We explore these representations by following the questions that matter in terms of explainability: *What* concepts does a vision encoder encode, and *where*, in which units and at what spatial resolution, does that information reside? Answering them without fine-tuning is attractive because it (a) avoids the cost and risk of retraining a large model, and (b) yields explanations about the *deployed* model rather than a modified copy. We therefore take a deliberately *training-free* approach. As shown in Figure 1, we freeze a 3D vision-language model and cache its final image embedding x_i . We then use multi-label volume annotations as statistical *probes* on x_i - not to train a network, but to ask: which embedding dimensions linearly encode clinical findings? This leads to three contributions: (1) **Sparsity** Each radiological finding is encoded by ~ 10 channels that achieve comparable classification performance to the full D-dimensional embedding, (2) **Causality** Zeroing a finding’s specific channels reduces its own probe AUROC score $\sim 20\times$ more than other findings, and the specific channels cluster into clinically coherent groups; and (3)

Generalization across backbones The sparse organization replicates on another 3D CT backbone (3D VLM with a different pretraining pipeline), indicating the phenomenon is a property of frozen VLMs rather than of one architecture.

We then show the probe is *useful* and *backbone-portable*: the same CCP-10 detections and corpus-derived template verbalizer outperform trained CT VLMs on clinical and factual metrics at a fraction of the cost on underlying 3D VLMs.

Related Work

3D medical vision-language models. Recent foundation models pair 3D CT encoders with text supervision, enabling zero-shot classification and report generation. Examples include CT-CHAT (Hamamci et al. 2026) and Pillar-0 (Agrawal et al. 2025) for chest CT, and Merlin (Blankemeier et al. 2024) for abdominal CT. These approaches optimize *what* the model predicts through pretraining and decoding. In contrast, we treat frozen encoders as interpretable systems and ask: *where* does finding information reside in the embeddings? We answer this through statistical probes rather than weight updates.

Probing and interpretability of learned representations. Linear probes (Alain and Bengio 2016) test what information is linearly decodable from frozen features, while network dissection (Bau et al. 2017) aligns individual units with human concepts. Superposition analyses (Elhage et al. 2022) show that features can be distributed across many coordinates, and representation-engineering methods (Zou et al. 2023) read and steer concepts via mean-difference directions. Our concept channel probe (CCP) adapts these ideas to 3D medical encoders: we rank channels by calibration-set selectivity and fit a closed-form mean-difference detector on a sparse top- K subset, yielding a per-finding, training-free probe.

Causal localization. Causal mediation analysis (Michel, Levy, and Neubig 2019) and activation-level interventions for locating and editing knowledge in networks (Meng et al. 2022) establish necessity by ablating internal components. We apply causal intervention to frozen embeddings: zeroing each finding’s top- K channels selectively impacts its score, leaving others unchanged.

Radiology report generation and evaluation. Report evaluation uses clinical metrics (RadGraph-F1 (Delbrouck et al. 2024), RadEval (Xu et al. 2025)). Rather than end-to-end generation, we decouple detection from verbalization: CCP

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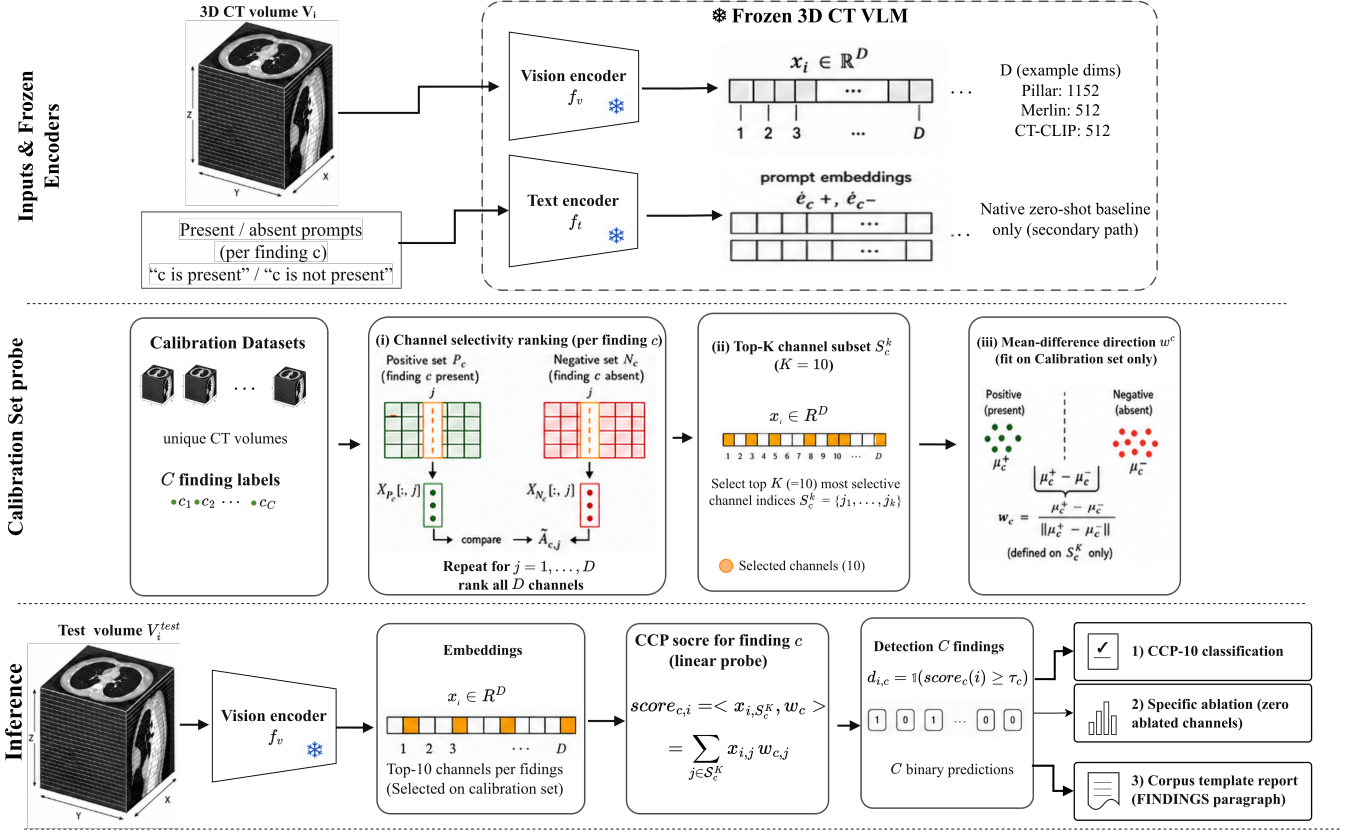


Figure 1: Concept Channel Probe (CCP) overview.

produces reliable binary detections that a deterministic corpus template turns into FINDINGS text, which we evaluate under the same clinical and surface metrics.

Method

We avoid weight fine-tuning and opaque generative decoding by taking a training-free approach: each 3D image passes through a *frozen* vision encoder to produce embedding x_i , on which all probes (CCP- K , sparsity, causal ablation, reports) operate without training.

Frozen Backbone Inputs

Let V_i denote a CT volume. Each backbone applies its own preprocessing; we cache the resulting frozen embedding per scan:

$$x_i = f_V(\text{preprocess}_V(V_i)) \in \mathbb{R}^D \quad (1)$$

where f_V and D are fixed by the pretrained model (e.g., PILLAR-0). We treat x_i as a black-box input; backbone internals remain frozen. Our channel-ranking and ablation logic generalizes across architectures; only D and preprocessing differ.

Concept Channel Probe (CCP- K)

CCP- K (*concept channel probe*) is our primary linear probe on frozen channels. Given a multi-label task with C findings,

for each finding $c \in \{1, \dots, C\}$, we (i) rank channel coordinates j by selectivity score $\tilde{A}_{c,j}$ on calibration set (Eq. 2) and form top- K sets S_c^K ; (ii) fit a closed-form mean-difference direction w_c on each S_c^K (Eq. 3); (iii) compute continuous scores $\text{score}_c(i)$, calibrate thresholds τ_c on a held-out *calibration* split, and design binary detections d_i (Eq. 4); and (iv) further analyze finding-specific channels and causal ablation (Eqs. 5). The calibration split is used for channel ranking (w_c), score calibration, and τ_c . Here the finding index is $c \in \{1, \dots, C\}$; channel index is $j \in \{1, \dots, D\}$; circuit size K is a hyperparameter fixed before evaluation. $\mathbb{I}(\cdot)$ is the indicator function (1 if the condition holds, 0 otherwise).

Channel ranking and scoring. For pathology c , let $P_c = \{i : \text{finding } c \text{ present}\}$ and $N_c = \{i : \text{finding } c \text{ absent}\}$ on the calibration set. We define per-coordinate selectivity following network dissection (Bau et al. 2017):

$$A_{c,j} = \frac{1}{|P_c||N_c|} \sum_{p \in P_c} \sum_{n \in N_c} \mathbb{I}(x_{p,j} > x_{n,j}), \quad \tilde{A}_{c,j} = A_{c,j} - \frac{1}{2} \quad (2)$$

$A_{c,j}$ is the pairwise win rate (equivalent to per-channel AUROC) and is invariant to finding prevalence—it measures separability, not class ratio. Centering to $\tilde{A}_{c,j}$ puts chance at zero. This makes ranking robust to the severe class imbalance in multi-label CT (prevalence ranges $\sim 7\%$ to $\sim 45\%$).

We rank channels by $\tilde{A}_{c,\cdot}$ and extract the top- K set $\mathcal{S}_c^K$, then fit a mean-difference projection following (Zou et al. 2023) to score unseen volumes as a sparse alternative to fine-tuning (Hamamci et al. 2026):

$$w_c = \underbrace{\frac{\mu_{\mathcal{S}_c^K}^P - \mu_{\mathcal{S}_c^K}^N}{\|\mu_{\mathcal{S}_c^K}^P - \mu_{\mathcal{S}_c^K}^N\|}}_{\text{fit on calibration set}}, \quad \underbrace{\text{score}_c(i) = \langle x_{i,\mathcal{S}_c^K}, w_c \rangle = \sum_{j \in \mathcal{S}_c^K} x_{i,j} w_{c,j}}_{\text{applied per volume}} \quad (3)$$

Here $\mu_{\mathcal{S}_c^K}^P$ and $\mu_{\mathcal{S}_c^K}^N$ are class means over calibration positives and negatives, restricted to top- K channels. This mean-difference approach avoids bias from class imbalance, unlike pooled or least-squares fits. Critically, $\mathcal{S}_c^K$ and w_c are computed only on the calibration set. At inference, both are frozen; scoring requires extracting $x_{i,\mathcal{S}_c^K}$ and computing its dot product with w_c . The test volume never enters μ^P or μ^N , ensuring no data leakage.

Binary detections. We turn each continuous probe score into a present/absent prediction by comparing $\text{score}_c(i)$ to a per-finding cutoff τ_c :

$$d_{i,c} = \mathbb{I}(\text{score}_c(i) \geq \tau_c), \quad d_i = (d_{i,1}, \dots, d_{i,C}) \in \{0, 1\}^C \quad (4)$$

where $d_{i,c}=1$ means CCP predicts finding c on volume i ; d_i is also the detection vector fed to the report generation process afterwards. On the calibration split we select each τ_c by sweeping candidate cutoffs on the calibration scores. We use two operating points depending on the downstream task: for the report-generation detections d_i we choose the cutoff that maximizes per-finding F1, whereas for the classification benchmark (Table 1) we follow the CT-CLIP protocol and place τ_c at the dev ROC point closest to the top-left corner—a prevalence-independent operating point that does not depend on the positive/negative ratio. In both cases we then freeze $\{\mathcal{S}_c^K, w_c, \tau_c\}_{c=1}^C$ and apply the same rule on the test split without refitting.

Causal ablation. The CCP- K classification selects the global top- K coordinates for each finding. To investigate whether these channels are *necessary* (i.e., causally responsible for diagnostic scores), we perform controlled ablation following Michel, Levy, and Neubig (2019) and Meng et al. (2022): for source finding c , define $\mathcal{S}_c^{\text{abl}}$ as the top- K coordinates with largest $\tilde{A}_{c,j}$ on the calibration split. We set $x'_{i,j}=0$ for $j \in \mathcal{S}_c^{\text{abl}}$ before recomputing frozen probe scores on a held-out split. The drop matrix is:

$$\Delta_{c \rightarrow c'} = \text{AUROC}_{c'}(\{x_i\}) - \text{AUROC}_{c'}(\{x'_i\}),$$

$$\text{sel} = \frac{1}{C} \sum_{c=1}^C \left(\Delta_{c \rightarrow c} - \frac{1}{C-1} \sum_{c' \neq c} \Delta_{c \rightarrow c'} \right) \quad (5)$$

where C is the number of findings. For each source c , the parenthesis in Eq. 5 is the gap between the *on-target* drop $\Delta_{c \rightarrow c}$ and the mean *off-target* drop over $c' \neq c$, and sel averages that gap over all c . Large sel means a finding’s specific channels are *necessary* for its own score but largely *unnecessary* for others.

Dataset	method	AUROC	F1	Accuracy	PR
CT-RATE ($n=1564$)	Random	0.500	0.566	0.507	0.189
	CT-CLIP zero-shot [§]	0.731	0.707	0.668	0.323
	CT-CLIP ClassFine [§]	0.756	0.724	0.689	0.339
	CT-CLIP VocabFine [§]	0.756	0.738	0.705	0.353
	PILLAR-0 ClassFine [†]	0.833	0.800	0.778	0.431
	PILLAR-0 zero-shot*	0.682	0.693	0.652	0.291
	CCP-Full — PILLAR-0	0.793	0.782	0.760	0.400
	CCP-10 — PILLAR-0 [‡]	0.798	0.790	0.768	0.405
RADCHEST-CT ($n=3630$)	Random	0.505	0.565	0.506	0.245
	CT-CLIP zero-shot [§]	0.629	0.637	0.592	0.335
	CT-CLIP ClassFine [§]	0.643	0.644	0.599	0.346
	CT-CLIP VocabFine [§]	0.650	0.677	0.636	0.346
	PILLAR-0 ClassFine [†]	0.713	0.609	0.570	0.335
	PILLAR-0 zero-shot*	0.634	0.579	0.533	0.295
	CCP-Full — PILLAR-0	0.683	0.596	0.553	0.324
	CCP-10 — PILLAR-0 [‡]	0.686	0.622	0.581	0.333

Table 1: Multi-abnormality classification (classification metrics: mean AUROC; mean per-finding weighted F1, accuracy, and precision (PR) after ROC upper-left thresholds on calibration set: CT-RATE official test set ($n=1564$) and RADCHEST-CT ($n=3630$). [§]CT-CLIP rows **extracted** from Hamamci et al. (2026); [†]Pillar ClassFine: trained on CT-RATE train labels ($\sim 39k$ volumes); encoder frozen. ***zero-shot**: Hamamci et al. (2026) present/absent prompt pairs ("c is present." / "c is not present.") embedded with each backbone’s text encoder, scored against cached x_i via softmax over two cosines—the same recipe as cited CT-CLIP ZERO-SHOT, applied to PILLAR-0 (1152-d) or Merlin (512-d, Clinical-Longformer); no label training. [‡]CCP-10: training-free channel probe + calibration set thresholds. **Bold** = best supervised per column; **bold underline** = best training-free per column.

Training-free report generation

We generate reports deterministically from d_i (Eq. 4) using a corpus-derived template. From training reports (excluding calibration examples), we extract: (i) T_0 , the most frequent FINDINGS report (a common "normal chest" skeleton); and (ii) $\{u_c\}$, the most common positive sentence per finding c with matching keywords. At test time, we edit T_0 : (1) split sentences, tag keywords, mark negation; (2) remove sentences for absent findings, append u_c for $d_{i,c} = 1$, generate IMPRESSION. The verbalizer never introduces findings beyond d_i (no hallucinated positives).

Experimental setup

Data. We evaluate on three public cohorts:

(1) **CT-RATE**: 3D chest CTs with radiology reports and $C = 18$ multi-label finding annotations. We use the official validation split (1564 CTs) for test evaluation and a 600-volume calibration set from *train* for channel ranking, probe fitting, and threshold calibration.

(2) **RADCHEST-CT** (Draeos et al. 2021): 3630 external chest CTs with 84 abnormality labels (52 location-based). Following the protocol of Hamamci et al. (2026), detectors

trained on CT-RATE dev are evaluated without refitting, using CT-RATE’s 18-finding ontology.

(3) MERLIN ABDOMINAL CT (Blankemeier et al. 2024): paired abdominal CTs with reports and $C = 30$ multi-label finding annotations.

Each finding is coded 1 (present), 0 (absent), or -1 (not assessed); we treat -1 as missing during channel ranking and probe fitting. Because prevalence is highly imbalanced across findings, we fit a detector for any finding with at least five positive calibration volumes and abstain otherwise; on CT-RATE all 18 findings qualify, while a few of the rarest RADCHEST-CT and MERLIN native labels fall below this floor.

3D VLM Backbones and K selection We run CCP on frozen precomputed vision-encoder embeddings from PILLAR-0 ($D = 1152$) and MERLIN ($D = 512$). On PILLAR-0 channel ranking and probe directions are fit on a 600-volume calibration set; K is chosen on a disjoint 800-volume train holdout by grid-searching $K \in \{1, 2, 5, 10, 20, 50, 100, 200, D\}$ and taking the smallest K within 0.005 mean AUROC of full- D ($K^* = 10$). We use the same $K^* = 10$ for MERLIN without separate tuning.

Classification metrics. F1, accuracy, and precision metrics follow CT-CLIP evaluation protocol (Hamamci et al. (2026)’s public evaluation code). Mean AUROC is threshold-free on raw scores. We report the numbers for CT-CLIP from the original paper since the public checkpoint is not available.

Report evaluation metrics. For report generation, we measure (i) **Clinical efficacy:** micro/macro-F1 from a RadBERT-CT labeler on generated text vs. official CT-RATE volume labels (same GT as CCP), (ii) **Surface NLG:** BLEU-1 and ROUGE-L vs. reference reports, (iii) **Factual structure:** RadGraph-F1 (Delbrouck et al. 2024), and (iv) **Reference-aligned:** RadEval (Xu et al. 2025) F1RadBERT-CT (Wald et al. 2025) on generated vs. reference report pairs.

Results

Multi-label classification

Table 1 summarizes results on CT-RATE and RADCHEST-CT. CCP uses frozen Pillar-0 embeddings. Baselines include: CT-CLIP zero-shot and ClassFine/VocabFine (cited from (Hamamci et al. 2026)); zero-shot prompts (" c is present/absent") scored via text-encoder similarity; and Pillar-0 ClassFine, a supervised linear head. CCP-10 is the best training-free method on both splits; supervised Pillar-0 ClassFine leads overall. On Pillar-0, training-free CCP-10 (0.798) approaches supervised ClassFine (0.833) without label training, while zero-shot stays lower (0.628–0.682 AUROC). On unseen RADCHEST-CT, CCP-10 remains best training-free (AUROC 0.686, F1 0.622), retaining $\sim 86\%$ in-domain AUROC vs. zero-shot. Supervised CT-CLIP VocabFine leads F1 (0.677); Pillar ClassFine reaches higher AUROC (0.713) but lower F1 (0.609).

Training-free report generation

We convert CCP-10 detections into reports on 1564 test volumes, holding detections fixed and varying only the *verbalizer*. Table 2 reports clinical efficacy and NLG metrics.

Method	Tr.	μ F1	M-F1	B1	R-L	Lt.
Structured template	no	.441	.408	.015	.064	.24
Frozen LLM	no	.315	.317	.057	.081	1.15
Corpus-based Template.	no	.549	.492	.483	.379	.24
Retrieval-edit	no	.540	.484	.439	.314	.24
k -NN retrieval	no	.544	.486	.414	.297	.24
CT-CHAT [†]	yes	.184	–	.373	.326	5.5
Radiologist	–	.962	.944	–	–	–

Table 2: Report generation on CT-RATE test ($n=1564$). Clinical Micor-F1 (μ F1), Macro-F1 (M-F1), evaluated by F1RadBERT-CT vs. official labels. NLG: BLEU-1 (B1) / ROUGE-L (R-L) vs. radiologist FINDINGS. Latency (Lt.) as wall-clock time to produce one report end-to-end based on second per volume (s/v). CCP-10 corpus-derived template verbalizer is based on PILLAR-0. [†]CT-CHAT: extracted from Hamamci et al. (2026).

We compare: **Structured template:** PRESENT/ABSENT lists formatted as FINDINGS/IMPRESSION. **Corpus-based template:** edits normal-study template T_0 with positive sentences $\{u_c\}$. **Frozen LLM:** off-the-shelf LLM (Qwen3-VL-8B-Instruct) generates reports from lists. **Retrieve-then-edit:** k-NN retrieves closest training report, then edits to match d_i . **k-NN constrained:** k-NN retrieval limited to normal reports, then edit to match d_i . **CT-CHAT:** trained chest CT VLM (Hamamci et al. 2026).

The CCP-10 corpus-based template is the strongest *training-free* verbalizer on all three axes: RadBERT-CT F1 0.549 micro / 0.492 macro (vs. cited CT-CHAT 0.184), BLEU-1 0.483 / ROUGE-L 0.379 (vs. CT-CHAT 0.373 / 0.326), and latency 0.24 s/vol— $\sim 23\times$ faster than CT-CHAT (5.5 s/vol). Retrieve-then-edit matches clinical F1 (0.540) but drops NLG (BLEU 0.439, ROUGE-L 0.314) due to non-corpus phrasing. Structured template and frozen LLM sacrifice prose style (BLEU ≤ 0.064 and ~ 0.01 respectively). Pairwise bootstrap confirms retrieve-then-edit and k-NN lag corpus F1 ($p < 0.001$), with corpus F1 in tight 95% CI (Table 3).

Statistical analysis. We assess uncertainty via paired volume bootstrap ($n = 1564$ test volumes, $B = 2000$ resamples; calibration/dev excluded). Point estimates report 95% bootstrap confidence intervals. Pairwise comparisons use two-sided bootstrap p -values with shared resampling indices. Classification uses mean macro-AUROC (threshold-free); report generation uses RadBERT-CT micro-F1 (Sec.). Table 3 summarizes primary comparisons.

Transfer across datasets, backbones, and anatomy

Since our concept channels probe (CCP) is defined entirely on frozen embeddings, the same recipe ($K = 10$) can be applied under three distinct transfer scenarios (Table 4): **(A) Cross-institution:** we use the same VLM backbone and chest anatomy, but different hospital/label ontology (PILLAR-0 on CT-RATE vs. RADCHEST-CT); **(B) Cross-anatomy/backbone:** we apply the same CCP recipe to another 3D VLM backbone as MERLIN on ABDOMINAL CT

Comparison	Δ	95% CI	p
<i>Report generation (Clin-F1 vs. corpus template)</i>			
Corpus template: 0.549 [0.537, 0.560] (absolute CI).			
Structured template	-0.078	[-0.086, -0.071]	$p < 0.001$
Retrieve-then-edit	-0.009	[-0.013, -0.004]	$p < 0.001$
k -NN retrieval	-0.005	[-0.007, -0.002]	$p < 0.001$
Frozen LLM (s2t)	-0.233	[-0.245, -0.221]	$p < 0.001$

Table 3: Paired bootstrap tests on CT-RATE test ($n = 1564$, $B = 2000$). Clin-F1: F1RadBERT-CT micro-F1; classification: mean macro-AUROC. Reference row for verbalizers: corpus template (CCP-10).

Classification transfer.					
Scce.	Calibration $\rightarrow$ Test	AUROC μ -F1			
<i>cross-institution (PILLAR-0 chest)</i>					
A	RADCHEST-CT valid $\rightarrow$ CT-RATE test	0.726	0.462		
A	CT-RATE dev $\rightarrow$ RADCHEST-CT all	0.686	0.622		
<i>cross-anatomy/backbone MERLIN</i>					
B	abd. CT valid $\rightarrow$ abd. CT test (30 findings)	0.839	0.832		
—	zero-shot [†]	—	0.647		
—	supervised [†]	—	0.641		
<i>anatomy mismatch, MERLIN encoder/Chest labels</i>					
C	CT-RATE train $\rightarrow$ CT-RATE test	0.733	0.733		
C	CT-RATE train $\rightarrow$ RADCHEST-CT	0.605	—		
Report generation transfer.					
Scce.	Calibration $\rightarrow$ Test	F1	RG-F1	B-1	R-L BERT
<i>cross-institution (PILLAR-0 chest)</i>					
A	RADCHEST-CT valid $\rightarrow$ CT-RATE test	0.448	0.284	0.471	0.352 —
<i>MERLIN abdomen</i>					
B	abd. valid $\rightarrow$ abd. test	—	0.262	0.214	0.226 0.677
—	Merlin+RadLlama [†]	—	0.293	0.102	— 0.588

Table 4: CCP-10 transfer summary. [†] extracted from Hamamci et al. (2026); Blankemeier et al. (2024)

dataset (Blankemeier et al. 2024); (C) **Anatomy mismatch**: we investigate whether the *chest* finding labels probed on an *abdominal*-pretrained MERLIN encoder can be applied to CT-RATE and RADCHEST-CT. For each scenario we rank channels, fit probe directions, and calibrate thresholds entirely on the *calibration* split of the source dataset, followed by evaluation on the *test* target data split. Report generation metrics utilize a fixed, corpus-derived template verbalizer, varying only the underlying detection source.

Scenario (A): PILLAR-0 cross-institution. We ask whether CCP probes *fit at one hospital* transfer to the other while keeping the frozen 3D VLM backbone (PILLAR-0) fixed. Probes are *fit* in each hospital’s native label space (18 on CT-RATE dev; 84 binary on RADCHEST-CT), but we *evaluate* on the target only through the mapped CT-RATE ontology similar to Hamamci et al. (2026). In **classification**, as shown in Table 4 (scenario A), the bidirectional transfer is asymmetric but strong RADCHEST-CT $\rightarrow$ CT-RATE test reaches AUROC 0.726 and F1 0.462; CT-RATE $\rightarrow$ RADCHEST-CT all volumes

reaches 0.686 / 0.622 (also reported in Table 1). In **report generation**, Table 4 (scenario A), we keep the CT-RATE corpus template and test split fixed and swap only the detection source to probes calibrated on RADCHEST-CT, which yields F1 0.448, BLEU 0.471, and ROUGE-L 0.352 vs. radiologist FINDINGS. It can be seen that without CT-RATE detector labels, cross-institution transfer retains $\sim 82\%$ of F1 in-domain report generation (see 2) and still exceeds CT-CHAT performance.

Scenario (B): MERLIN cross-anatomy/backbone We apply CCP-10 to abdominal CT classification by ranking sparse channels in frozen Merlin contrastive embeddings, and calibrating per-finding probe directions and thresholds on the valid split with 30 findings, and evaluating the frozen channels on the test set. As shown in Table 4 (scenario A), in **classification** task, CCP-10 reaches AUROC 0.839 and F1 0.832, exceeding MERLIN zero-shot and supervised (Blankemeier et al. 2024) 0.647 and 0.641 F1, respectively. In **report generation** (Table 4, scenario B): with the same CCP-10 detections, we mine a corpus template from Merlin *train* FINDINGS, apply the same deterministic edit rule. Our proposed method yields RadGraph-F1 (RG-F1) 0.262, BLEU-1 (B1) 0.214, ROUGE-L (R-L) 0.226, and BERTScore F1 (BERT) 0.677. Compared to Merlin+RadLlama (Blankemeier et al. 2024), our training-free method improves BLEU-1 and BERTScore (0.214 and 0.677 vs. 0.102 and 0.588) but yields lower RadGraph-F1 (0.262 vs. 0.293).

Regime (C): Abdominal encoder MERLIN on chest Labels, anatomy mismatch We apply the same CCP-10 recipe to CT-RATE chest findings but swap the frozen encoder to abdominal-pretrained MERLIN—an intentional anatomy mismatch that complements Blankemeier et al. (2024)’s external benchmarks, where matched-anatomy zero-shot F1 drops from 0.741 to 0.647 under hospital *distribution shift* yet frozen MERLIN still generalizes to external chest CT in their linear-probe evaluation (Blankemeier et al. 2024). Probes rank channels, fit directions, and calibrate thresholds on CT-RATE dev, then evaluate frozen readouts on official CT-RATE test ($n=1564$) and on all RADCHEST-CT volumes ($n=3630$; mapped ontology). In **classification** (Table 4, regime C), we can see on CT-RATE test, CCP-10 reaches AUROC 0.733 and F1 0.733; near original CT-CLIP zero-shot (0.731 / 0.707; but below chest-native PILLAR-0 CCP-10 (0.798 / 0.790). Cross-institution evaluation on RADCHEST-CT all volumes yields AUROC 0.605, well below regime (A) PILLAR-0 transfer (0.686 / 0.622). Regime (C) therefore complements MERLIN’s external chest benchmarks (Blankemeier et al. 2024), and we observe that the CCP procedure transfers across backbones, but encoder pretraining and probe labels must align, as shown in Scenarios A and B.

Analysis and discussion

CCP-10 succeeds on frozen chest and abdominal encoders. Here we analyze *where* that signal lives in the embedding. We ask: (i) Are findings encoded by sparse channels? (ii) Are these channels causally necessary? (iii) Does this hold across anatomies?

Each finding lives in ~ 10 channels. We rank channels by per-concept AUROC on the calibration set, apply CCP with

varying top- K , and evaluate on 1564 test volumes at $K = 10$. Fig. 2 shows mean test AUROC across 18 CT-RATE findings vs. top- K channels on PILLAR-0. Performance rises sharply from top-1 (0.752) to top-10 (0.798), saturates by top-50 (0.804), and reaches 0.793 with all $D = 1152$ coordinates. Top-10 matches full-dimension AUROC (0.798 vs. 0.793) using only 10 coordinates per finding ($\sim 100\times$ fewer), and beats zero-shot text prompting (0.682). Top-10 can exceed full-embedding AUROC on individual findings (e.g., lung nodule 0.734 vs. 0.683, atelectasis 0.746 vs. 0.697) by dropping noisy dimensions. Table 5 lists each finding with its channel set $\mathcal{S}_c^{10}$ and test AUROC. Gaps are modest in both directions ($|\Delta \text{AUROC}| \leq 0.05$ for all 18 labels). Hiatal hernia shows the largest gap (top-10: 0.027 below full). Fluid findings stay near ceiling (pleural effusion 0.928/0.944, consolidation 0.905/0.886).

Causal ablation: Do sparse channels matter? We show that a few channels ($K = 10$) correlate strongly with each finding; however, correlation alone does not establish that the selected coordinates are necessary for discrimination. On CT-RATE test set (PILLAR-0, $n=1564$), we zero $\mathcal{S}_c^{\text{abl}} = \mathcal{S}_c^K$ (Eqs. 3, 5) for each source finding c and record $\Delta_{c \rightarrow c'}$ for every scored finding c' . Figure 3 displays the resulting 18×18 matrix with source c on rows and scored c' on columns (Eq. 5). We ablate the same dev-ranked top- K channels that CCP uses for each finding ($\mathcal{S}_c^{\text{abl}} = \mathcal{S}_c^K$). If those coordinates encoded only generic shared signal, zeroing finding c 's circuit would lower every detector by about the same amount; instead, the mean on-target AUROC drop is 0.0060 versus 0.00003 off-target ($\sim 20\times$), so each finding's channels are *selectively necessary* for its own score rather than broadly shared across all labels. Some off-diagonal entries remain elevated among clinically related findings (e.g., arterial and coronary calcification; emphysema with atelectasis and lung opacity), consistent with partial circuit overlap among related pathologies.

Merlin circuits by clinical category. We aggregate the top-10 CCP circuits into 11 finding categories, following Blankemeier et al. (2024). Fig. 4). Category-level selectivity (max over findings of $\tilde{A}_{c,j} = \text{AUROC}_{c,j} - 0.5$ per coordinate) localizes sparse channels to clinically meaningful anatomy (e.g., lower-thorax calcification and cardiac findings share coordinates, whereas pleural effusion uses a separate circuit; peritoneal ascites and free air reach mean top-10 AUROC 0.895. Vasculature findings, by contrast, show heterogeneous top-10 sets, consistent with diverse vascular appearances. Cross-category Jaccard overlap is near zero for unrelated groups (liver vs. lower thorax) but rises where pre-training anatomy overlaps (pancreas/spleen vs. lower thorax, 0.43), mirroring the partial circuit sharing among related chest findings in Fig. 3. Together, these results suggest contrastive 3D pretraining concentrates finding information in sparse coordinates rather than spreading it uniformly. CCP localizes, thresholds, and ablates these coordinates without training.

Interpretability vs. supervised finetuning. Supervised finetuning (CT-CLIP ClassFine/VocabFine, Pillar ClassFine; Table 1) optimizes predictions by updating decision boundaries. CCP localizes *where* information lives by ranking chan-

Finding	$\mathcal{S}_c^{10}$	top-10	all D
medical material	525, 923, 941, 904, 589, 1110, 848, 429, 870, 457	0.889	0.849
arterial wall calcification	792, 897, 1145, 564, 588, 607, 490, 981, 889, 97	0.864	0.863
cardiomegaly	897, 1103, 1145, 813, 423, 607, 792, 1045, 761, 429	0.878	0.863
pericardial effusion	552, 627, 761, 570, 1087, 923, 560, 872, 609, 446	0.813	0.767
coronary artery wall calcification	792, 1145, 588, 564, 490, 607, 897, 889, 407, 981	0.879	0.866
hiatal hernia	1078, 875, 607, 981, 1126, 457, 521, 588, 490, 407	0.652	0.679
lymphadenopathy	897, 1137, 1045, 1072, 813, 549, 985, 604, 634, 901	0.713	0.721
emphysema	771, 1100, 1009, 385, 792, 1107, 716, 429, 1069, 545	0.760	0.770
atelectasis	751, 584, 586, 623, 483, 613, 851, 968, 1039, 967	0.746	0.697
lung nodule	643, 508, 661, 663, 557, 386, 462, 528, 442, 543	0.734	0.683
lung opacity	404, 595, 721, 719, 517, 648, 743, 707, 594, 602	0.856	0.858
pulmonary fibrotic sequela	559, 586, 578, 613, 751, 454, 680, 1066, 727, 1064	0.682	0.692
pleural effusion	570, 609, 446, 395, 552, 761, 498, 659, 560, 444	0.928	0.944
mosaic attenuation pattern	897, 1103, 1007, 879, 858, 813, 784, 894, 982, 1045	0.792	0.786
peribronchial thickening	885, 564, 239, 1069, 429, 771, 457, 484, 904, 792	0.717	0.744
consolidation	689, 426, 648, 413, 551, 549, 600, 901, 730, 418	0.905	0.886
bronchiectasis	781, 716, 454, 1107, 1000, 961, 545, 397, 1100, 577	0.727	0.753
interlobular septal thickening	743, 1097, 1072, 428, 740, 482, 1013, 927, 719, 721	0.823	0.861

Table 5: Per-finding CCP- K^* circuits on PILLAR-0 (CT-RATE validation, $n=1564$, $D=1152$). For each finding, $\mathcal{S}_c^{K^*}$ lists dev-ranked channel indices (Eq. 3); top- K^* and all- D columns are threshold-free test ROC-AUC.

nels and fitting a closed-form probe on frozen weights. On CT-RATE, CCP-10 exceeds reported CT-CLIP zero-shot and remains the best training-free probe, uniquely enabling sparsity and ablation analysis. On RadChest-CT, CCP-10 leads AUROC but lags VocabFine on weighted F1. Finetuning optimizes scores; CCP audits deployed encoders and provides channel-level explanations alongside predictions.

Detection-grounded reports. We separate detection (CCP) from verbalization, avoiding end-to-end VLM failure modes. CT-CHAT achieves BLEU-1 0.373 but F1 only 0.184, drifting toward normal wording and missing abnormalities. CCP-10 + corpus template wins on both metrics (F1 0.549, BLEU-1 0.483) at $\sim 23\times$ lower latency. Two checks support this design. Unrelated reports score BLEU-1 0.385 against each other, so BLEU reflects boilerplate, not correctness. Corpus templates preserve institutional negations better than frozen LLMs. Retrieve-then-edit lags corpus template on F1, showing that with trustworthy detections, the bottleneck is faithful reconciliation to d_i , not generation. On abdominal CT, CCP-10 + Merlin template improve NLG scores over Merlin+RadLlama without fine-tuning. Trained RadLlama achieves higher RadGraph-F1 through richer entity encoding.

Transfer across institutions, backbones, and anatomy. The three regimes in Table 4 form a single generalization story: Regime (A): Sparse chest circuits from one hospital retain signal at another (AUROC 0.686 / 0.726; report

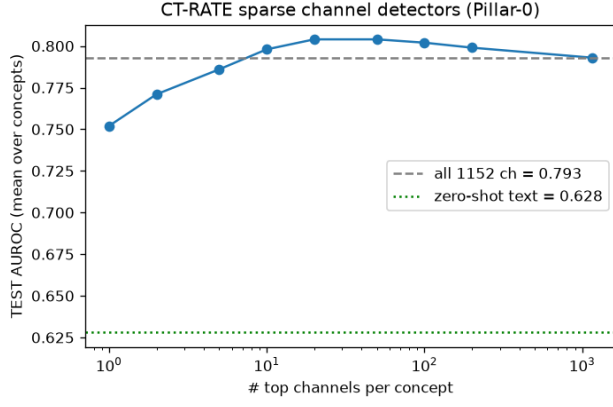


Figure 2: Sparsity recovery: test classification AUROC vs. number of top channels per finding. A handful of channels recover most of the full-feature performance on PILLAR-0.

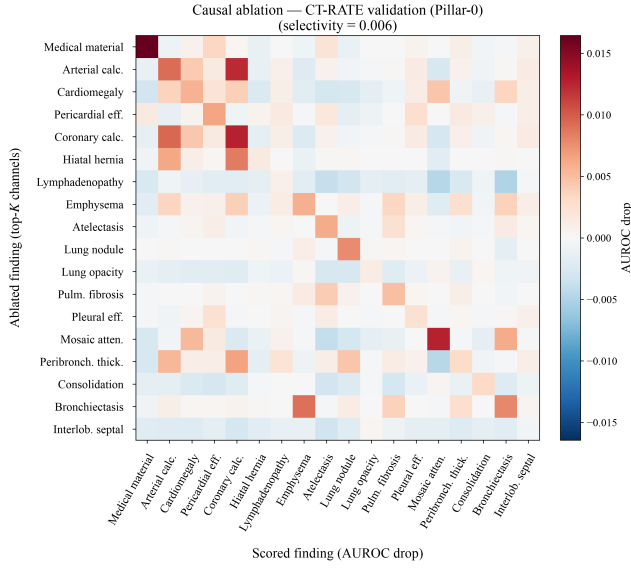


Figure 3: Causal ablation on CT-RATE validation with frozen PILLAR-0. Row c : AUROC changes after zeroing source finding c 's dev-ranked top- $K=10$ circuit ($S_c^{\text{abl}} = S_c^K$); column c' : scored finding. Cell (c, c') shows $\Delta_{c \rightarrow c'}$ from Eq. 5. A strong diagonal indicates selective necessity; off-diagonal blocks indicate shared circuits among related findings.

F1 $\sim 82\%$ of in-domain). Institutional distribution shift does not erase channel structure. Regime (B): The CCP recipe is backbone-portable. On abdominal MERLIN, CCP exceeds Merlin zero-shot and supervised F1 (0.832 vs. 0.647 / 0.641). Regime (C): Chest probes on abdominal MERLIN remain in-domain (0.733) but lag chest-native Pillar-0 and degrade under hospital shift (0.605 vs. 0.686). Encoder and probe anatomy must align; the ranking procedure transfers across backbones, not across mismatched anatomies.

This work presents one consistent picture: Frozen 3D CT encoders store findings in sparse causally selective channel subsets. CCP localizes and names these coordinates training-

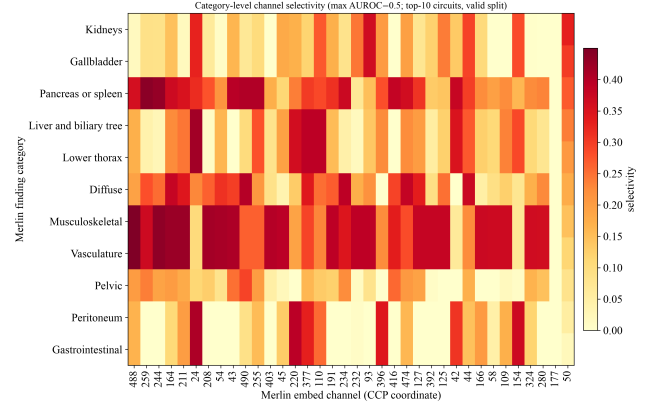


Figure 4: **Merlin abdominal CCP circuits by finding category (Figure 2c grouping).** Each row is one of Merlin’s 11 clinical categories (30 findings total). Columns show embed coordinates with highest category-level selectivity (max AUROC=0.5 across findings in that category; valid split). Color intensity = selectivity.

free, whereas finetuning improves *what* is predicted but not *where* it is encoded. CCP-10 achieves competitive performance while providing interpretability, grounding language in reliable detections rather than free-form generation. The probe-and-readout recipe transfers across hospitals and backbones when anatomy matches, and breaks down when it does not (regime C). For deployed foundation models, AUROC alone is an incomplete account of trust. Users need assurance that findings rest on stable internal features. CCP audits deployed models by probing, naming, ablating, and verbalizing. A limitation: rare findings have higher-variance channel estimates (few positive calibration volumes). We mitigate by reporting threshold-free AUROC, using prevalence-independent ROC operating points, and excluding low-count findings. Whether channel-level explanations remain stable under finer pathology labels, longitudinal follow-up, and multimodal inputs remains future work.

Conclusion

In this work we show how frozen medical VLMs encode radiological findings in sparse, causally selective channels. CCP localizes, ablates, and converts these into predictions and reports on chest (PILLAR-0) and abdominal (MERLIN) settings. The probe-and-template pipeline is competitive with finetuned VLMs on classification and report metrics while operating at a fraction of the latency. CCP can serve as a fast perception module in multi-agent workflows, supplying structured detections and channel evidence to routing agents, reducing latency in multi-step pipelines. We view CCP as a practical route to auditing *what* and *where* encoders represent findings, enabling lightweight, repeated inference without full VLM cost. Future work should test whether channel circuits remain stable under finer pathology labels, longitudinal follow-up, and multimodal inputs. We will also evaluate CCP inside multi-agent benchmarks with latency and interpretability as first-class metrics.

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